

The multiplicity- m theorem for an arbitrary coordinate order

Claude, with Daniel Murfet

2026-09-07

Abstract

Relabelling the coordinates of the box along an equivalence $\tau : \text{Fin } D' \simeq \text{Fin } D$ leaves the chart integral invariant when ξ, η are transported along the induced measurable equivalence. Sorting the exponents puts the minimisers last, so unit 80 yields the general form of the leading term of `thm:TaylorTree`: for any exponents on $\text{Fin } D$ whose minimal candidate exponent p is attained $M + 2 \geq 2$ times, and jointly continuous ξ, η with $\eta > 0$ on the minimal face, $\int_{(0,b]^D} w^H \eta e^{-\beta n w^{2K} + \beta \sqrt{n} w^K \xi} \sim C n^{-p/2} \log^{M+1} n$ with $C > 0$. Zero `sorry/axiom`.

1 The things formalised

```
noncomputable def relabel {D D' : } ( : Fin D' → Fin D) : (Fin D' → ) → (Fin D → )
theorem boxIntegralGen_relabel ( b c : ) {D D' : } ( : Fin D' → Fin D) (K H : Fin D → )
  ( : (Fin D → ) → ) :
  boxIntegralGen b c (K ) (H ) ( relabel ) ( relabel )
  = boxIntegralGen b c K H
theorem boxIntegralGen_isEquivalent_general ( b : ) (h : 0 < ) (hb : 0 < b) {D : }
  (K H : Fin D → ) (hK : i, 0 < K i) (p : ) (hmin : i, p ≤ finExp K H i) (M : )
  (hM : (Finset.univ.filter fun i => finExp K H i = p).card = M + 2)
  ( : (Fin D → ) → ) (h c : Continuous ) (h c : Continuous )
  (hpos : w : Fin D → , ( i, finExp K H i = p → w i = 0) →
    ( i, finExp K H i ≤ p → 0 < w i ≤ b) → 0 < w) :
  C : , 0 < C (fun n : => boxIntegralGen b (Real.sqrt n) K H ) ~[atTop]
  fun n : => C * (n ^ (-(p / 2))) * Real.log n ^ (M + 1))
```

2 Proof strategy

Relabelling invariance is `measurePreserving_piCongrLeft` applied to `MeasurePreserving.integral_comp'`, with the monomials reindexed by `Fintype.prod_equiv`. For the general theorem, $\sigma = \text{sortPerm}$ makes the exponents antitone; the minimisers form the final segment $\text{Ici } i_0$ of cardinality $M + 2$, so $i_0 = D - (M + 2) =: d'$, and $\tau = \text{finCongr} \circ \sigma$ relabels $\text{Fin}(d' + (M + 2))$. The exponent hypotheses of unit 80 follow from the segment description; the face condition transports through `relabel_apply_apply` and `Fin.addCases` on $\tau^{-1}i$.

3 Roadblocks and resolutions

- `MeasurableEquiv.piCongrLeft_apply_apply` needs the family given explicitly (`( := fun _ => )`) when the equivalence is wrapped in a definition.

- `Finset.card_equiv` maps the left set into the right one: with the sorted set on the right the equivalence is σ^{-1} , not σ .
- `Fin.addCases` does not generalise hypotheses mentioning the eliminated index; **revert** them first.

4 Outcome

The leading term of `thm:TaylorTree` is formalised for general continuous ξ, η on the genuine chart box in any coordinate order and any multiplicity ($m = 1$ by unit 72, $m \geq 2$ here). Next: consult Astra on the remainder and expansion programme (targets 6–8).